

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CT-1 EXAMINATION • AY 2025-26

Course: Computer Networks (CS503) | Faculty: Prof. Sunita Rao

Date: 05 October 2025

Time: 75 Minutes

Max Marks: 25

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — PHYSICAL & LINK LAYER ANALYSIS

Answer both questions.

1. An IP packet with length 4000 bytes arrives at a router with MTU of 1500 bytes. Show all fragments with offset, MF flag, and length. [12 M]
2. Calculate the CRC remainder for message $M = 1101011011$ using generator polynomial $G(x) = x^4 + x + 1$. [13 M]